Taken from "The Global Brain"
by Peter Russell

Chapter 1

The Blue Pearl

> Once a photograph of the Earth, taken from the outside, is available . . . a new idea as powerful as any in history will let loose.
>
> *Sir Fred Hoyle, 1948*

What is home? To a man visiting a neighbor across the road, home is his house, his rose bushes, his backyard. To the peasant bringing goods to the city, it is his village. To the traveler abroad, it is his country. These are common human experiences. At one time or another, we each have called our town, state, or nation "home." Yet there is a greater home that only recently has come into our awareness, even though it has been ours all along: planet Earth.

As the first astronauts traveled out into space and the Earth receded into the distance, national boundaries began to lose their significance. These space pioneers no longer found themselves identifying with a particular country, class, or race but with humanity and the planet as a whole. Standing on the lunar surface, the astronauts saw what no human being had ever seen before: the great sphere that is Earth, four times as big and five times as bright as the moon itself.

To Edgar Mitchell, the sixth man to stand on the moon, this was a deeply moving experience, and he felt a strong connection to the planet. "It was a beautiful, harmonious, peaceful-looking planet, blue with white clouds, and one that gave you a deep sense . . . of home, of being, of identity. It is what I prefer to call instant global consciousness."

Mitchell observed that everyone who has been to the moon has had similar experiences. "Each man comes back with a feeling that he is no longer an American citizen—he is a planetary citizen."

Russell Schweickart, another astronaut, similarly felt a profound change in his relationship to the planet.

> You realize that on that small spot, that little blue and white thing, is everything that means anything to you—all of history and music and poetry and art and death and birth and love, tears, joy, games—all of it on that little spot out there. . . . You recognize that you are a piece of this total life. . . . And when you come back there is a difference in that world now. There is a difference in that relationship between you and that planet and you and all those other forms of life on that planet, because you've had that kind of experience.

But the astronauts were not the only ones to have experienced this profound change of perspective. The photographs of our planet brought back from space triggered similar deep reactions in many Earth-bound men and women, feelings of awe and connectedness.

The profound impact of this Earth view has resulted in this picture being used in almost every sphere of human activity. It adorns the walls of offices and living rooms; it is a greeting card, a T-shirt, and a bumper sticker. Ecological movements and planetary organizations incorporate it in their logos, as do educational institutions and business corporations. At one time or another it has been used to advertise just about everything from cars, washing machines, and shoes, to book clubs, banks, and insurance companies. Yet in spite of all this exposure, the picture still strikes a very deep chord, and none of its magnificence has been lost.

It is not entirely coincidental that this photograph achieved its widespread appeal at the same time that many people were becoming increasingly concerned about the relationship between humanity and the planet, and our need to live in harmony with each other and with our environment. The picture has become a spiritual symbol for our times. It stands for the growing awareness that we and the planet are all part of a single system, that we can no longer divorce ourselves from the whole.

So the most valuable spinoff from the moon expeditions may have been, not in the fields of science, economics, politics, or the military, but in the field of consciousness. Getting to the moon allowed humanity for the first time to look upon this blue pearl that has been our home for millions of years and to see it as a whole. As Edgar Mitchell pointed out, "The payoff from Apollo may be inestimably richer than anyone anticipated."

The Living Earth

The view of Earth from space brought with it yet another insight: the possibility that the planet as a whole could be a living being. We Earthlings might be likened to fleas who spend their whole lives on an elephant, unaware of what it really is. They chart its terrain—skin, hairs, and bumps—study its chemistry, plot its temperature changes, and classify the other animals that share its world, arriving at a reasonable perception of where they live. Then one day a few of the fleas take a huge leap and look at the elephant from a distance of a hundred feet. Suddenly it dawns: "The whole thing is alive!" This is the truly awesome realization brought about by the trip to the moon. The whole planet appears to be alive—not just teeming with life but an organism in its own right.

If the idea of the Earth as a *living being* is initially difficult to accept, it may be due partly to our assumptions about what sort of things can and cannot be organisms. We accept a vast range of systems as living organisms, from bacteria to blue whales, but when it comes to the whole planet we might find this concept a bit difficult to grasp. Yet until the development of the microscope less than four hundred years ago, few people realized that there are living organisms within us and around us, so small that they cannot be seen with the naked eye. Today we are viewing life from the other direction, through the "macroscope" of the Earth view, and we are beginning to surmise that something as vast as our planet could also be a living organism.

This hypothesis is all the more difficult to accept because the living Earth is not an organism we can observe ordinarily outside ourselves; it is an organism of which we are an intimate part. Only when we step into space can we begin to see it as a separate being. Stuck like fleas on an elephant, we have not, until recently, had the chance to see the planet as a whole. Would a cell in our own bodies, seeing only its neighboring cells for a short period, ever guess that the whole body is a living being in its own right?

To better understand the planet as a living system, we need to go beyond the time scales of human life to the planet's own time scale, vastly greater than our own. Looked at in this way, the rhythm of day and night might be the pulse of the planet, one full cycle for every hundred thousand human heartbeats. Speeding up time appropriately, we would see the atmosphere and ocean currents swirling round the planet, circulating nutrients and carrying away waste products, much as the blood circulates nutrients and carries away waste in our own bodies.

Speeding it up a hundred million more times, we would see the vast continents sliding around, bumping into each other, pushing up great mountain chains where they collided. Fine, threadlike rivers would swing first one way then another, developing huge, meandering loops as they accommodated themselves to the changes in the land. Giant forests and grasslands would move across the continents, sometimes thrusting limbs into new fertile lands and at other times withdrawing as climate and soil changed.

If we could look inside, we would see an enormous churning current of liquid rock flowing back and forth between the center of the planet and the thin crust, sometimes oozing through volcanic pores to supply the minerals essential for life.

Had we senses able to detect charged particles, we would see the planet bathing not only in the light and heat of the sun but also in a solar wind of ions streaming from the sun. This wind, flowing round the Earth, would be shaped by her magnetic field into a huge, pulsating aura streaming off into space behind her for millions of miles. Changes in the Earth's fluctuating magnetic state would be visible as ripples and colors in this vast cometlike aura, and the Earth herself would be but a small blue-green sphere at the head of this vast energy field.

Thus if we look at the planet in terms of its own time scales, we seem to see a level of complex activity similar to that found in a living system. Such similarities, however, do not constitute any form of proof. The question we have to ask is whether scientists could accept the planet as a single organism in the same way they accept bacteria and whales? Could the Earth actually "be" a living organism?

This no longer seems so farfetched. On the contrary, an increasingly popular scientific hypothesis suggests that the most satisfactory way of understanding the planet's chemistry, ecology, and biology is to view the planet as a single living system.

The Gaia Hypothesis

One of the major proponents of the theory that the planet behaves like a living system is British chemist and inventor Dr. James Lovelock. His ideas, which have fundamentally altered many people's perception of the planet, were another fortuitous spinoff from the space race.

In the early 1960s Lovelock served as consultant to a team at the California Institute of Technology working on plans for the investigation of life on Mars. One problem they faced in looking for Martian life-forms was not knowing exactly what they were looking for. Other life-forms might be based on completely different chemistries—on silicon rather than carbon, for instance—and might not reveal themselves to tests based on Earthly life.

Lovelock theorized that however strange the chemistry and life-form might be, there would be one very general characteristic: any life-form would take in, process, and cast out matter and energy, and this would have detectable effects upon its physical surroundings. On a planet devoid of life, the chemical constituents of its atmosphere, oceans, and soil, through their interactions over millions of years, would settle into equilibrium; the proportions of the various constituents could be predicted roughly by the laws of physical chemistry. If, however, life were present, then whatever chemical processes it was based on almost certainly would leave the environment in a state recognizably different from that predicted by physical chemistry alone.

As a very simple example of this principle, consider a jar containing a mixture of sugar and water. Physical chemistry predicts that the sugar will dissolve until a given concentration is reached. If life in the form of yeast cells were added and left to grow, however, the resulting mix would be very different: there would be a lower concentration of sugar than predicted and much higher levels of alcohol and other organic products. We would determine, then, whether there was (or had been) life in the jar by measuring the sugar and alcohol concentrations.

The beauty of Lovelock's approach to life detection is that one need not visit another planet to know whether or not life is there. The basic chemistry of the atmosphere can be deduced from Earth-bound examination of the infrared, light, and radio waves coming from the planet. In the 1960s enough was known about the Martian atmosphere to suggest that it was very close to the state of chemical equilibrium; it showed no signs of the exotic chemistry characteristic of the presence of life. So, Lovelock concluded it was extremely unlikely that there was life on Mars.

Applying a similar approach to the atmosphere, oceans, and soil of our own planet, Lovelock found that the chemical constituents were far removed from equilibrium. To the casual observer it might seem that he had merely shown that there was, after all, life on Earth. But Lovelock began to see far greater significance in these disequilibria.

First, the concentration of gases in the Earth's atmosphere differs by factors of millions from the levels predicted by physical chemistry. The predicted level of oxygen in the air, for example, would be virtually zero, yet the actual concentration is about 21 percent. This is puzzling because oxygen is a highly reactive gas, combining readily with many other chemical elements; it should therefore be rapidly absorbed. Second, and even more puzzling, the actual composition of the atmosphere has remained at a level that is optimal for the continuance of life.

After pondering many such unlikely characteristics, Lovelock came to "the only feasible explanation": the atmosphere is being manipulated on a day-to-day basis by the many living processes on Earth. The entire range of living matter on Earth, from viruses to whales, from algae to oaks, plus the air, the oceans, and the land surface all appear to be part of a giant system able to regulate the temperature and the composition of the air, sea, and soil so as to ensure the survival of life. This concept Lovelock termed the "Gaia Hypothesis" in honor of the ancient Greek "Earth Mother," *Gaia* (or *Ge*). In this context Gaia signifies the entire biosphere—everything living on the planet—plus the atmosphere, the oceans, and the soil.

In maintaining the optimal conditions for life, Gaia manifests a characteristic that all living systems have in common: homeostasis. Derived from the Greek for "to keep the same," the term was coined by Claude Bernard, a nineteenth-century French physiologist, who stated that "all the vital mechanisms, varied as they are, have only one object: that of preserving constant the conditions of life."

An example of homeostasis is the human body's maintenance of a temperature of about 98.6 degrees Fahrenheit, the ideal temperature for the majority of the body's metabolic processes. Although the external temperature may vary by scores of degrees, our internal temperature seldom varies by more than a degree or two, the body cooling itself through sweating and warming itself through physical activity and shivering. The regulation of the number of white blood cells, and the control of the acidity, salt content, and delicate chemical balance of the blood are homeostatic processes as well. These and many others together maintain the best internal environment for the continuance of our body's life processes. Such processes are found not only in the human body and in all living systems but also within Gaia herself.

Gaia appears to maintain planetary homeostasis in a variety of ways, monitoring and modifying many key components in the atmosphere, oceans, and soil. The data that Lovelock amassed in support of this contention is fascinating, and the interested reader should take a look at Lovelock's book *Gaia: A New Look at Life on Earth*. In summary, some of the indications of Gaia's homeostatic mechanisms are:

The steadiness of the Earth's surface temperature: Although life is found to exist between the extremes of 20 and 220 degrees Fahrenheit, the optimal range is between 60 and 100 degrees Fahrenheit. The average temperature of most of the Earth's surface appears to have stayed within this range for hundreds of millions of years despite drastic changes in atmospheric composition and a large increase in the heat received from the sun. If at any time in the Earth's history the overall temperature had gone beyond these limits, life, as we know it, would have been extinguished.

The regulation of the amount of salt in the oceans: At present the oceans contain about 3.4 percent salt, and geological evidence shows that this figure has remained relatively constant, despite the fact that salt is being washed in continually by the rivers. If the salt concentration had ever risen as high as 4 percent, life in the sea would have evolved very differently. If it had exceeded 6 percent, even for a few minutes, life in the oceans immediately would have come to an end, for at this level of salinity cell walls disintegrate. The oceans would have become like the Dead Sea.

The stabilization of the oxygen concentration of the atmosphere at 21 percent: This is the optimal balance for the maintenance of life. With a few percent less oxygen, the larger animals and flying insects could not have found enough energy to survive; with a few percent more, even damp vegetation would burn. (A forest fire started by lightning would burn fiercely and indefinitely, eventually burning all vegetation on the Earth's land surface.)

The presence of a small quantity of ammonia in the atmosphere: This is the amount needed to neutralize the strong sulphuric and nitric acids produced by the natural combination of sulphur and nitrogen compounds with oxygen (thunderstorms, for example, produce tons of nitric acid). The result is that rain and soil remain at the optimal levels of acidity for the preservation of life.

The existence of the ozone layer in the upper atmosphere: This shields life on the surface from ultraviolet radiation, which damages the molecules essential for life, particularly the DNA molecules found in every living cell. Without it life would have been very rapidly annihilated.

On the basis of these and other "homeostatic" behaviors, Lovelock concludes that the climate and chemical properties of the Earth seem always to have been optimal for life as we know it.

Critics of the Gaia Hypothesis might argue that the origin and maintenance of life on this planet has resulted from a series of very lucky coincidences, rather than planetary homeostatic behavior. If, for example, the proportion of ammonia in the early atmosphere had been a little higher or lower, the Earth would have ended up too hot or too cold for life. They might argue that it has been a series of flukes that kept the planet's surface temperature roughly constant while the sun's output changed; a series of flukes that kept the levels of carbon dioxide, oxygen, salt, and many other chemicals at optimal levels for the maintenance of life; and a fluke that there is an ozone layer to protect us from lethal quantities of ultraviolet light.

In the same way, a cell in the human body, observing the body's continued survival through heat, cold, and many other changes, might, if it were so inclined, put it all down to a series of lucky coincidences; the body just happens to sweat when it is hot, just happens to shiver when it is cold, just happens to take in the right amount of nutrients when they are needed. Perhaps by fluke blood sugar, acidity, and salinity stay at the optimal levels and red blood cells happen to bring along oxygen and take away wastes. From such a point of view, the body survives from one moment to the next as a result of an extremely fortunate series of coincidences.

This quite definitely is not the case. The body behaves in a well-ordered manner with a definite sense of purpose. It sweats, shivers, eats, breathes, and regulates its internal functions and chemical constituents in order to preserve homeostasis, and so survive.

Just as this self-regulating principle makes more sense of the body's activities, so it makes more sense of the planet's. Gaia appears to be a self-regulating, self-sustaining system, continually adjusting its chemical, physical, and biological processes in order to support life and its continuing evolution.

Does the Gaia Hypothesis imply that the biosphere is a single living organism? Lovelock is cautious on this point. He sees the atmosphere to be similar to a beehive, a biological construction designed to maintain a chosen environment, though not actually living in itself. But if we take the atmosphere, oceans, and soil to be intrinsic parts of a complete biosystem, couldn't we then speculate that the system as a whole is alive? And if so, is humanity an intricate, inseparable part of a larger living system? Before we can answer these questions, we need to look more closely at the general characteristics common to all living systems and see to what extent Gaia satisfies them.